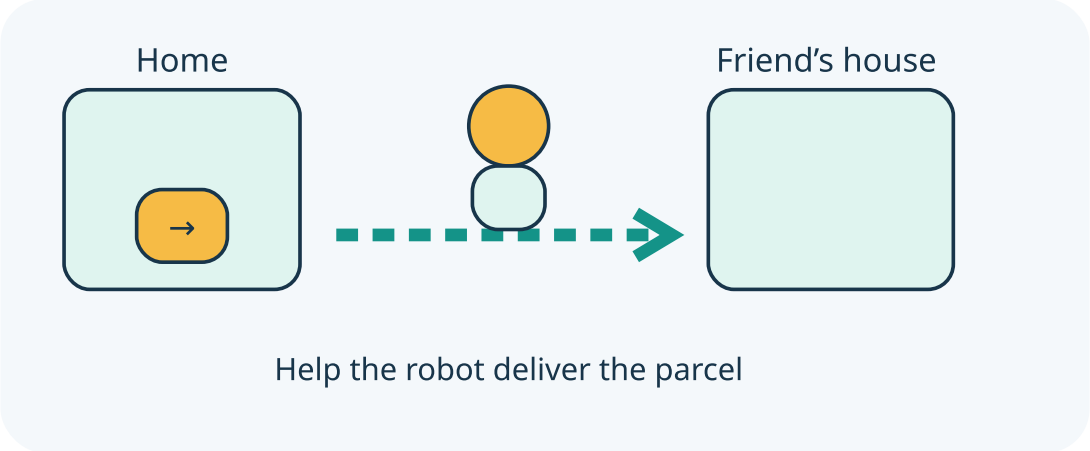
Debugging Strategy

Parcel delivery

Rule: check the name before picking up the box. Broken plan: deliver → walk → pick up → check name. Reorder it so every step can work.



Answer: Check the name on the box → Pick up the box → Go to the friend’s house → Hand over the box

Check the name → pick up the box → go to the house → hand it over. The order gets the right parcel to its owner.

Think further

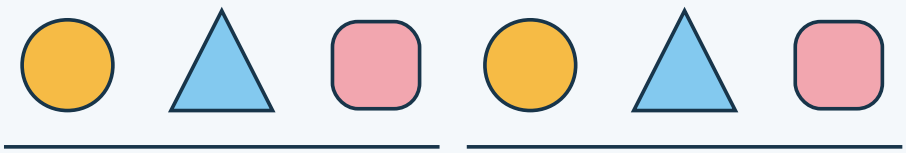
What could happen if the name is not checked?

It may go to the wrong person, so follow the name-check rule.

Functions

Call a bead recipe

The command “Three-shape set” makes circle → triangle → square. Two calls make this picture. What does one more call add?



“Three-shape set” called twice

Answer: Circle • triangle • square

Calling the same recipe adds circle → triangle → square again, without listing each shape again.

Think further

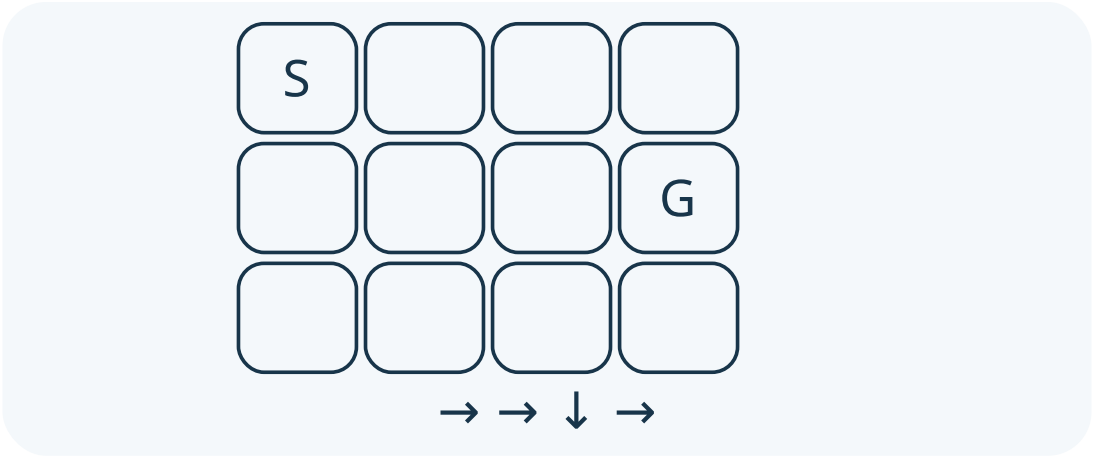
Draw two repeats of your own pattern.

Accept any clear named pattern repeated consistently twice.

Variables & Data

Count delivery steps

The counter starts at 0. Each arrow adds 1. The robot follows → → ↓ → to G. What is the counter now?



Answer: 4 steps

The counter changes: 0 → 1 → 2 → 3 → 4. It stores the changing number of steps.

Think further

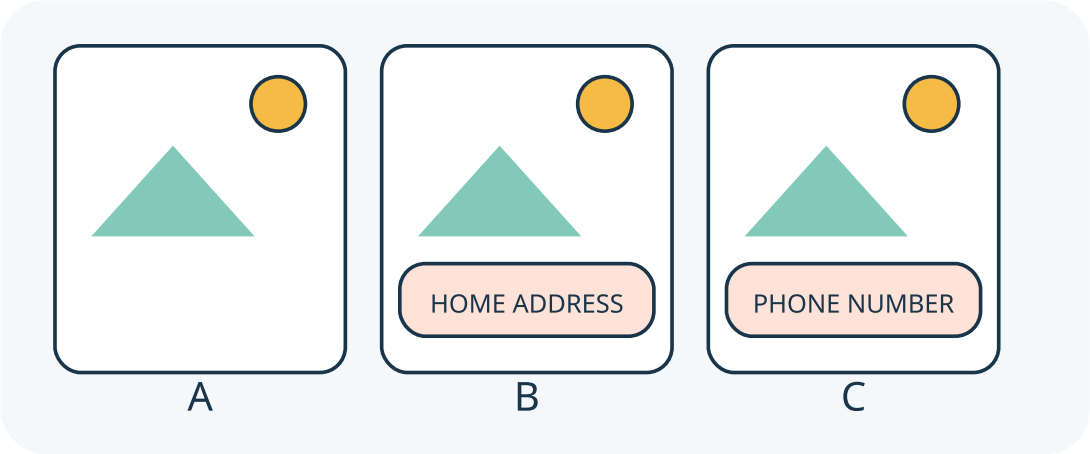
If the robot takes 2 more steps, what is the counter?

6 steps, because 4 + 2 = 6.

Digital Footprint + Cybersecurity Basics

Which picture is safer to share?

You want to post your drawing publicly. Which picture avoids revealing private details?



Answer: Only the drawing, with no identifying details

Choose only the artwork. An address or phone number is not needed to show it. Ask a trusted adult before posting.

Think further

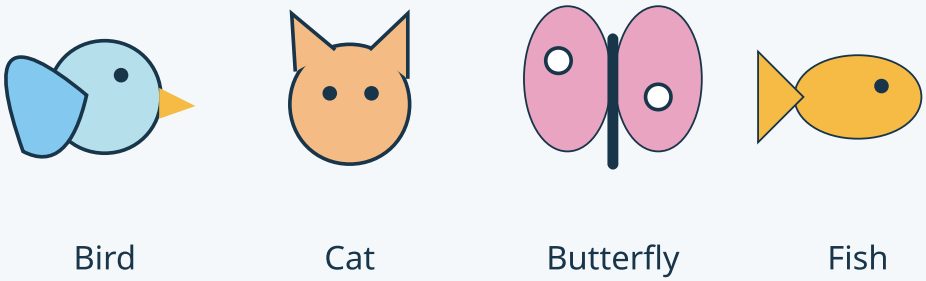
If a friend is in the photo, what should you do before sharing?

Ask the friend and a trusted adult, or do not share the picture.

Nested Logic

Robot animal gate

If it has wings: if the wings have spots → A, otherwise → B. If it has no wings → C. Use the drawings.



Bird Cat Butterfly Fish

Answer: Bird: B Plain wings • Cat: C No wings • Butterfly: A Spotted wings • Fish: C No wings

The drawn bird has plain wings: B. The butterfly has spotted wings: A. Cat and fish have no wings: C. The second question is inside the first.

Think further

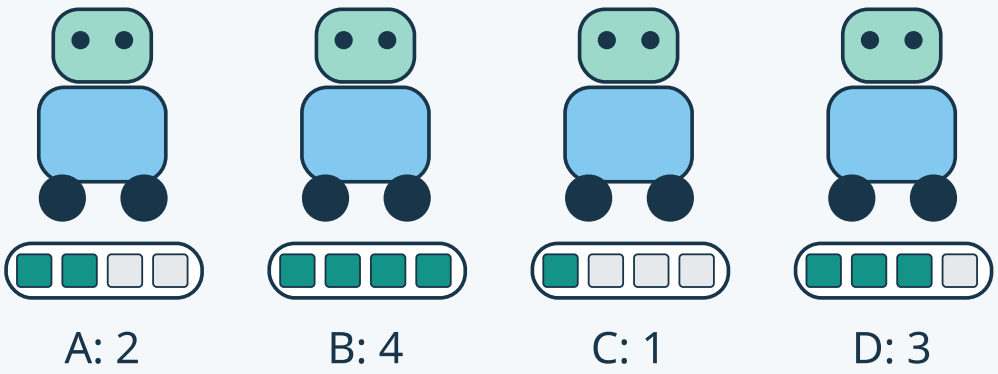
Draw an imaginary animal that can use gate A.

The drawing must have wings with spots, matching the rule.

Variables & Data

Which robots can deliver?

In this game, one delivery needs 3 energy units. A robot needs at least 3 units. Select every ready robot: A has 2, B has 4, C has 1, D has 3 units. These numbers are game rules, not real battery facts.



Answer: B: 4 units / D: 3 units

B has 4 and D has 3, enough for the mission. A has 2 and C has 1, which are not enough. We use the current energy values to decide.

Think further

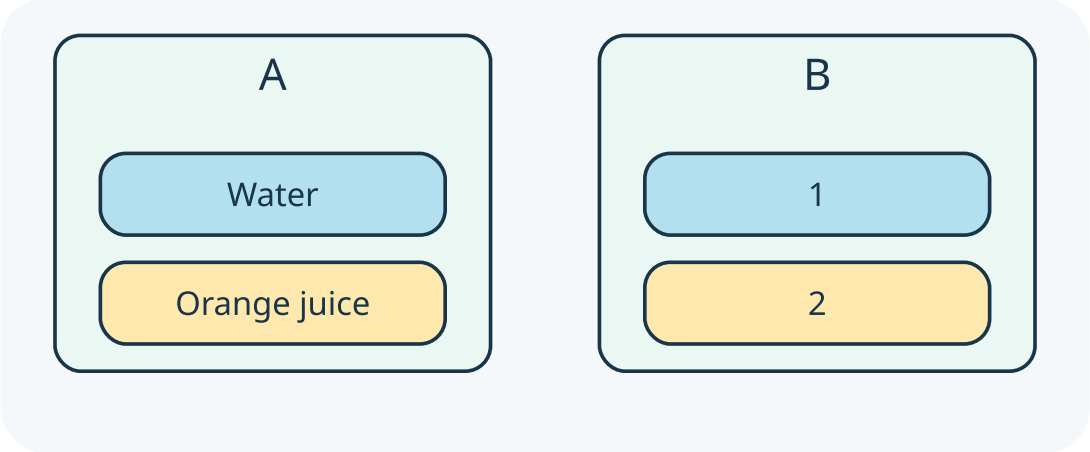
If A gains 1 energy unit, which robots are ready now?

A, B and D. A changes from 2 to 3; B and D keep their values.

Simple UI

Drink machine

A first-time user wants water. Which design helps them choose without guessing?



Answer: A: buttons labelled “Water” and “Orange juice”

A names the drink directly. Numbers alone need an explanation. Attractive colours do not automatically make a design clear.

Think further

Draw a clear button for choosing milk.

Use a clear “Milk” label. A milk picture may help; colour alone is not enough.

Simple UI

Did my order go through?

After tapping Order, what should the screen show to confirm the order?



Answer: Rice order received • Queue 12

Confirmation names the order and queue. It reduces guessing and repeated ordering.

Think further

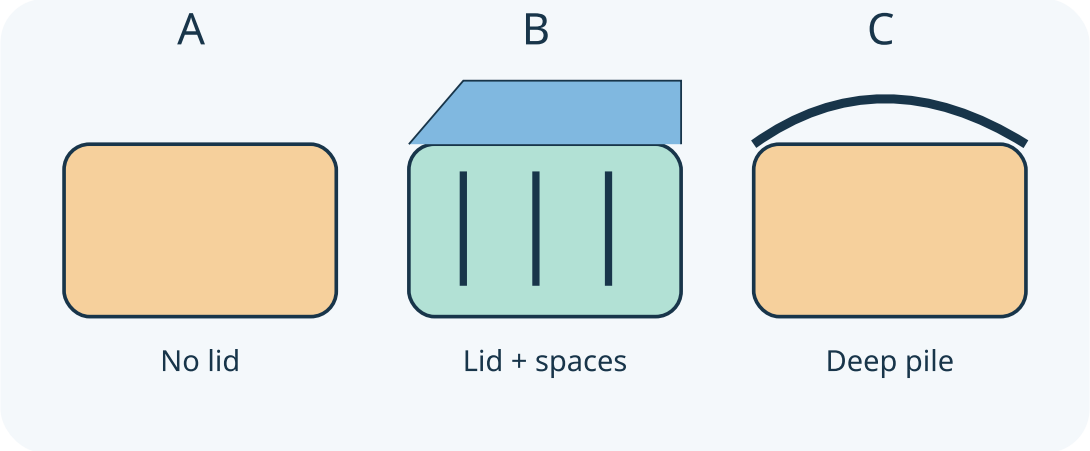
If the order fails, what should the screen say next?

For example: “Order not received. Please try again.” Do not claim success.

Prototype & Test

Min’s pencil box

Min tested the designs: A does not close, easy to reach; B closes, easy to reach; C closes, hard to reach. Min needs both. Which fits the test results?



Answer: B Box with lid and pencil spaces

Test results show B meets both needs. Use what Min tried, not your favourite colour. If real use reveals a problem, change it and test again.

Think further

Draw your own box to meet Min’s needs. Show it to a friend and ask, “What is unclear?” Choose one thing to improve.

Answers vary. Show how the box closes and allows easy access. Name something unclear to the friend and a planned improvement. Looking at a drawing is not a test of a real box.